Gloxad Academy: what has been done

Progress update, 11 November 2026. Covers the web app (Gloxadapp) and the backend (gloxard-backend).

1. Summary

- **New: a Practice area** in the student dashboard. Learners write and run JavaScript, Python, or HTML/CSS/JS in the browser, with live preview and automatic saving to their account. The backend needed new code for this and now has it, with 32 passing checks.
- **Free courses now go module by module.** Lessons, then a 5-question quiz (70% to pass), then a written task, then the next module. The rules are enforced on the server, not just on screen.
- **Trimester and scholarships** have public landing pages and a full backend. The home page Trimester section was removed at your request; the Trimester pages remain.
- **Provider credit and categories** now follow the new backend data: shared provider records, category tags, pathways and hide/show.
- **A crash found and fixed:** the live API now sends a course category as an object, which blanked free-course pages. Fixed in five places.
- **Site polish:** header, search, profile menu, toasts, course cards, skeleton loaders, mobile menu, hero and settings page were all reworked and checked on desktop and phone.
- **Not yet live:** the Trimester, scholarship, coding-lab and Practice backend routes are not on the Render server yet (see section 2).

2. What is live and what is not

Checked against https://gloxard-backend.onrender.com on 11 November 2026 with read-only requests.

Area	On the live server	Notes
Module-by-module free courses (Part A)	Live	Courses carry sequentialModules and the module and task routes exist.
Provider registry and category tags, pathways, aliases	Live	/api/providers answers. It holds no providers yet, and every course has provider: null.
Public Trimester list and detail	Not live	Returns 404. Needs the backend deployed.
Scholarships (public and admin)	Not live	Returns 404.
Coding labs	Not live	Returns 404.
Practice workspaces (new)	Not live	Exists in the local backend only. The Practice page falls back to browser-only scratchpads until it is deployed.
22-node curriculum trail	Not confirmed	The docs still show the old weekly shape. No example response was available, so the screen is not built.

3. Work completed

3.1 Practice area (new)

Part	What it does
Languages	JavaScript, Python (real Python in the browser, downloaded once, about 10 MB) and HTML/CSS/JS with a live preview that updates as you type.

Part	What it does
Running code	Code runs only in the learner's browser, in a sandbox. Endless loops are stopped (4 s for JavaScript, 10 s for Python). Web pages run in a locked iframe that cannot reach the app. Console output and errors show in a panel.
Saving	Saves in the browser on every keystroke and to the account about 2 seconds after typing stops. Status shows Saved, Saving, Offline or Changed elsewhere.
Safe with several devices	If another device saved first, a banner offers Load the latest or Keep mine. Nothing is overwritten silently.
History	Earlier versions can be listed and restored. Restoring keeps the current code in history too.
Fallback	Without a working practice API (demo accounts, or the Render server before it is updated) the page offers three scratchpads kept in the browser and says so.
Where to find it	Practice in the student sidebar and the top navigation, at /dashboard/practice. Works on phone width.
Backend	New endpoints under /api/practice/workspaces: list, create, open, autosave, versions, restore, delete. Only the owner can see a workspace. Limits: 10 files, 100,000 characters per file, 30 workspaces per user. The server stores code but does not run it.

3.2 Free courses: module by module

Part	What it does
Order	Lessons, then module quiz (5 questions, 70% to pass), then a written task (at least 50 words), then the next module opens.
Server rules	403 MODULE_LOCKED, 409 LESSONS_INCOMPLETE and 409 QUIZ_NOT_PASSED. The browser cannot skip ahead.
Quizzes and tasks	Generated for each module with stable ids, so a catalogue re-import does not break progress. Correct answers are only returned after an attempt.
Screens	Locked-module badges, a module task form with word count, and quiz result handling in the course player.

3.3 Trimester and scholarships

Part	What it does
Pages	/trimester shows a programme landing page (or a card list when there are several), a four-week journey, and a button matched to the viewer: Sign in to enrol, Enrol now, Continue learning, Opens on, Closed. /scholarships lists upcoming, open and closed scholarships with an apply form and application status.
Backend	Scholarship and application models, public lists, enrol, apply, admin create, edit, publish, archive and review. Application history is only ever added to.
Home page	The scholarships banner now shows real published scholarships. The home Trimester section was removed on request.

3.4 Provider credit and categories (from the Phase 1 backend work)

Part	What it does
Credit fix	Free courses now read the API's provider and source fields. Before, live courses showed a generic YouTube creator label. They now show the real creator, such as freeCodeCamp.org, with the right channel link.

Part	What it does
Provider card	A course with a provider record shows a Content provider card with its ownership class (external, Gloxad-produced or licensed) and a link to the original. No live course has a provider yet, so this card has not been seen working.
Providers service	New service for /api/providers and /api/providers/:idOrSlug.
Categories	Categories carry tags, pathways, aliases and active state. Public pages hide inactive ones. The admin screen has a show/hide toggle, a Hidden badge and a pathways line. The toggle was not tested against the live server (no admin login).
Crash fixed	The live API sends a course category as an object (id, name, slug). The free course page, recommendations, home explore tabs, admin course table and tutor course list rendered it as text and crashed. One shared helper now handles both shapes.

3.5 Catalogue

The free catalogue holds 80 courses. Six were added from providers other than freeCodeCamp so the site does not lean on one source. Licence terms for providers such as non-commercial Creative Commons still need confirming by whoever handles licensing.

3.6 Site polish and responsiveness

Area	What changed
Header and search	Cleaner navbar, a search with an icon button, tidier social icons, a modern profile menu and logout icon.
Course cards	One fluid card that fits any width, with skeleton loaders while data loads. Home page lists shortened and use different providers.
Mobile	A cleaner slide-out menu, fixed text touching images in hero and section layouts, and no sideways scroll at 375 px on the pages checked.
Home page	More hero text and room for the photo grid, richer scholarships banner, tidier testimonials.
Account screens	Sign in and sign up buttons refined; settings page, change-password card and profile menu restyled.
Toasts	Cleaner, modern alert and sign-in messages.
Institution and Tutor pages	Icons, social icons and hero text fit fixed.

4. Test results

Check	Result
Practice API (scripts/test-practice.js)	32 of 32 passed: ownership, conflicts, safe retries, limits, history, restore, delete.
Module gating (scripts/test-module-gating.js)	27 of 27 passed.
Trimester and LMS (scripts/test-lms.js)	117 passed on a clean rebuilt database.
Regression (scripts/test-unit12.js)	19 of 19 passed.
Type check (tsc)	Clean after the Practice, provider and category changes.
Practice in the browser	JavaScript and Python output, Python errors, a Python endless loop stopped at 10 s, web preview and console, autosave surviving a reload, the conflict banner and Load the latest, history list, phone layout and the browser-only fallback.
Course page on live data	Renders again; creator shows as freeCodeCamp.org with the channel link.
Not tested	Restore button in the browser (covered by the API tests), the category show/hide toggle, the provider card, and skeleton loaders outside the Trimester list.

5. Decisions and open items

- **Deploy the backend** to Render and run npm run catalog:import. Until then the Trimester, scholarship, coding-lab and Practice features cannot work on the live site.
- **22-node curriculum trail:** send an example response of GET /api/lms/cohorts/:cohortId/curriculum and the screen can be built.
- **Providers that do not allow embedding:** the spec says those lessons should open at the source. Decide how such lessons are marked complete, since progress depends on the embedded player.
- **Workspace path:** the spec names /api/lms/workspaces/:id/autosave. The Practice API is at /api/practice/workspaces. Decide whether to add the cohort-linked path as well.
- **No server-side code running:** results are unverified. The spec requires an independent security review before untrusted code runs on the server.
- **Licences:** confirm terms for catalogue providers.
- **Commit:** the Practice, provider, category and crash-fix changes are not committed yet.

6. Next

- Add and rename files in Practice, and link the existing coding-lab challenges into the Practice page.
- Provider admin screens and a provider filter on the course lists, once providers are filled in.
- The curriculum trail screen, once its response shape is known.
- Assignments, grading, final project and instructor and admin tools from the Weekly Developer Guide.